

Chethan Chakradhar Mangala

Senior Data Scientist / AI/ML Engineer

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PROFESSIONAL SUMMARY:

- **Over 10+ years of experience** in **Artificial Intelligence (AI)**, **Machine Learning (ML)**, **Natural Language Processing (NLP)**, and **Generative AI (Gen AI)** with extensive hands-on work across structured and unstructured datasets, predictive modeling, data analysis, data acquisition, data validation, and data visualization in diverse enterprise environments.
- Hands-on experience designing and deploying LLM-powered applications using LLaMA 3.1, GPT (OpenAI), and BERT, with specialization in Conversational AI, Retrieval-Augmented Generation (RAG) pipelines, and Multilingual NLP systems.
- Built and deployed advanced AI chatbots using OpenAI APIs, Hugging Face Transformers, and Google Vertex AI, driving improvements in customer support automation and engagement.
- Familiar with Transformer-based models such as BERT, GPT-3/4, and domain-tuned variants, performing real-time NER, Sentiment Analysis, and Document Classification across multiple languages.
- Proficient in Python and core ML libraries: NumPy, Pandas, matplotlib, Seaborn, Sci-kit Learn, SciPy, Keras, TensorFlow 2.x, PyTorch 1.x/2.x, NLTK, and spaCy.
- Hands-on with ML algorithms including Linear/Logistic Regression, Naïve Bayes, Random Forest, Decision Tree, SVM, KNN, Multiple Regression, XGBoost, K-Means, PCA, and Ensemble Methods.
- Built deep learning models using ANN, CNN, RNN, LSTM, Encoders, GANs, and Transformers, optimized for production deployment using TensorFlow and PyTorch.
- Monitored model performance using observability and monitoring tools, tracking accuracy, drift, bias, and data quality metrics.
- Designed and operationalized scalable ML pipelines on AWS services including SageMaker, Lambda, EC2, EMR, Step Functions, CloudWatch, Secrets Manager, S3, and Redshift.
- Developed ML APIs with Flask, integrated with Azure ML Studio, Google Cloud AI Platform, and accelerated training with GPU optimization strategies.
- Applied Time Series Analysis techniques such as ARIMA, LMS, Filtering, and Neural Networks for forecasting, anomaly detection, and signal modeling.
- Strong hands-on with Big Data technologies: Hadoop, HDFS, Hive, HBase, Spark, PySpark, Spark SQL, Kafka, and Sqoop for high-volume data pipelines and ETL.
- Designed analytical systems using Star Schema, Snowflake Schema, and ODS architecture, ensuring scalable and maintainable Data Warehousing infrastructure.
- Experienced in writing complex queries using SQL (PostgreSQL, SQL Server, MySQL) and working with NoSQL databases like MongoDB, with deep understanding of query tuning and data modeling.
- Built Entity Resolution pipelines for deduplication, record linking, name disambiguation, and NER matching using ML and fuzzy logic.
- Proficient in unsupervised learning including K-Means, Hierarchical Clustering, DBN, LDA, and PCA for dimensionality reduction, topic modelling, and segmentation.
- Implemented supervised learning models such as Decision Trees, Support Vector Machines, Bayesian Networks, and Neural Networks for classification, regression, and ranking tasks.
- Automated ETL workflows and report generation using Python, SQL, and Apache Airflow, and visualized outputs via Tableau, Power BI, SAS Visual Analytics, and D3.js.
- Performed statistical modelling and text analytics using tools like R (ggplot2, dplyr, caret), SPSS Modeler, SAS, and WEKA in complex predictive analytics projects.
- Built and optimized ML pipelines for production using Anaconda, Jupyter Notebook 6.x, and RStudio, enabling collaborative experimentation and reproducibility.

Strong object-oriented programming skills with practical application of design patterns, versioned code management using Git, and CI/CD using GitHub Actions in Agile/Scrum environments.

TECHNICAL SKILLS:

GenAI&AI solutions	LangChain, Snowflake GenAI, LLM Fine-tuning, OpenAI, Vertex AI, Hugging Face, Retrieval-Augmented Generation (RAG), Multimodal LLMs, ChromaDB, CoreML, BERT, GPT models, Text-to-SQL, Custom Embedding Models, Vector Search Optimization, Prompt Engineering, Agentic AI
Model development	Natural Language Processing (NLP), Machine Learning (ML), Deep Learning, Large Language Models (LLMs), Chatbots.
Machine Learning techniques	Linear Regression, Logistic Regression, Naive Bayes, Decision Trees, Random Forest, SVM, K-Means, KNN, Gradient Boosting Trees, AdaBoost, PCA, LDA, NLP Techniques.
ML frameworks & Libraries	Scikit-learn, TensorFlow, Keras, PyTorch, Pandas, NumPy
Deep Learning Expertise	Expertise in Neural Networks, CNNs, RNNs, LSTMs, Autoencoders, GANs, Transformer Architectures, Multimodal LLMs.
NLP Frameworks and Algorithms	NLTK, SpaCy, BERT, GPT, NER, Sentiment Analysis, Tokenization, Text Classification, Vector Embedding Models.
BigData Framework	Apache Spark, Apache Hadoop, Kafka, Databricks, Unity Catalog, Oracle Cloud Infrastructure (OCI), IBM DB2, Snowflake, Elasticsearch, GCP Analytics
DevOps/DataOpsTools	Jenkins, Docker, Terraform, OpenShift, Bitbucket CI/CD, AWS Lambda, Kubernetes Service Debugging, Build/Release Automation.
Programming Languages	Python, R, Scala, Java, SQL, TensorFlow, PyTorch, Shell Scripting, Terraform, Docker, Kubernetes, Flask, FastAPI.
Data Processing &Visualization	Data Cleaning, Feature Scaling, Data Imputation, Data Augmentation, MLflow, SQL, Hive, Athena, Tableau, Seaborn, Matplotlib, Plotly, Dash.
Web Development & Databases	Flask, Django, HTML5, JavaScript, Ajax, jQuery, Oracle, MySQL, PostgreSQL, MongoDB, Cassandra, Neo4j, AWS Neptune, Vector Stores, Expertise in Query Optimization, Graph Traversal, Schema Design.

PROFESSIONAL EXPERIENCE:

Client: Verizon, USA

June 2024 – Present

Role: Sr Data Science / Sr AI/ML Engineer

Responsibilities:

- Architected and maintained scalable data pipelines using AWS S3, Glue, and Redshift to integrate large-scale customer interaction and campaign data across distributed marketing environments.
- Migrated legacy batch analytics workflows to AWS EMR using PySpark, enhancing data availability and reducing processing time for timely campaign insights.
- Designed and maintained end-to-end ML pipelines using MLflow, Kubeflow, and Vertex AI, enabling automated training, validation, deployment, and retraining workflows
- Designed and orchestrated end-to-end ETL workflows using Apache Airflow (MWAA) and AWS Step Functions to automate complex dependencies between marketing data sources and analytics models.

- Developed distributed data processing solutions using Spark 3.x on AWS EMR, enabling high-throughput analytics across structured CRM data and unstructured persona interview transcripts.
- Built data analytics solutions using Amazon Redshift, Athena, and Spark SQL to uncover account engagement patterns, buyer sentiment trends, and campaign performance bottlenecks.
- Implemented CI/CD pipelines using AWS Code Pipeline, Code Build, and Code Deploy to streamline deployment of data processing modules supporting creative AI and chatbot systems.
- Ensured compliance with Model Risk Management (MRM) standards, including documentation, explainability, validation, and audit readiness.
- Leveraged Vertex AI AutoML and H2O Driverless AI for rapid model development, documentation automation, and accelerated deployment cycles.
- Created interactive executive dashboards in Amazon QuickSight for real-time monitoring of key B2B marketing KPIs such as account activity, lead generation, and content engagement.
- Automated data transformations in PySpark to standardize multi-source qualitative and quantitative data supporting intent classification and segmentation.
- Optimized inference performance through container tuning, autoscaling, and caching strategies.
- Automated model lifecycle processes including feature ingestion, training, testing, versioning, and promotion across environments.
- Implemented materialized views and clustering keys in Snowflake to accelerate complex query performance for large-scale persona and customer data analytics.
- Enabled seamless data ingestion into Snowflake from cloud sources (AWS S3, Azure Blob) leveraging Snowpipe and external stages for continuous data flow.
- Engineered modular, reusable data models and schema layers to support enterprise-wide account-based marketing analytics and AI-powered personalization.
- Ensured data security and compliance by integrating data governance including access control, lineage tracking, and audit logging in analytics pipelines.
- Implemented CI/CD pipelines for ML models integrating code, data, and model versioning, ensuring reproducible and auditable deployments.
- Integrated Snowflake with Python, dbt, and Apache Airflow to automate end-to-end data workflows feeding advanced AI models for buyer intent analysis.
- Led zero-downtime cloud migration initiatives using AWS Lambda, S3, and Redshift Spectrum to enable phased ingestion and processing of marketing data streams.
- Enabled automated rollback, retraining triggers, and environment parity across dev, test, and production.
- Integrated Git-based version control using AWS CodeCommit to support agile collaboration and change tracking across data engineering teams.
- Configured AWS CloudWatch and Cost Explorer to monitor system health and optimize compute/storage costs across multi-environment marketing analytics deployments.
- Established real-time monitoring systems to detect ETL failures and latency issues, ensuring timely delivery of data for AI model training and campaign execution.
- Supported secure data sharing using Snowflake's Data Sharing feature to enable cross-departmental collaboration on customer insights and campaign data.
- Authored detailed technical documentation covering data infrastructure, pipeline workflows, and schema designs for audit readiness and team onboarding.
- Collaborated cross-functionally with marketing strategists, data scientists, and product managers to align data solutions with evolving go-to-market and AI objectives.

Environment: AWS S3, AWS Glue, Amazon Redshift, Amazon Athena, AWS EMR, PySpark, Spark SQL, Apache Spark, AWS Lambda, Apache Airflow (MWAA), AWS Step Functions, Amazon QuickSight, Python, SQL, ETL, Data Modelling

Client: BCBS, USA

Oct 2022-May 2024

Role: Sr Data Science / Sr AI/ML Engineer

Responsibilities:

- Developed and deployed deep learning models using TensorFlow, PyTorch, and Keras for equipment performance forecasting, anomaly detection in machinery operation, and NLP-based maintenance ticket classification, improving downtime reduction and operational efficiency at Caterpillar.
- Applied transformer architectures and RNNs for time-series modeling to enhance machine usage prediction and improve workload balancing across manufacturing and construction equipment.
- Built distributed data pipelines with Apache Spark (3.x), Scala, and SQL optimized for telemetry ingestion from heavy machinery, sensor data analysis, and real-time usage pattern detection.
- Automated ETL workflows using Apache Airflow (2.x) and Prefect, integrating multi-source telemetry (IoT sensors, SCADA data) and operational logs for unified equipment analytics.
- Fine-tuned large language models to develop industrial-specific AI assistants for predictive maintenance and operator support.
- Explored Vision-Language Models (CLIP, BLIP) for processing image-based equipment inspection reports and automated defect recognition in manufacturing.
- Deployed containerized ML workloads using Docker, Kubernetes, and CI/CD tools integrated with Bitbucket Pipelines enabling scalable deployment across cloud and on-premise industrial environments.
- Deployed and operated ML workloads on AWS, GCP, and Azure, leveraging Docker and Kubernetes for scalable and resilient infrastructure.
- Leveraged AWS, Azure, and GCP cloud services for hybrid deployment and federated training using SageMaker, Vertex AI, and Azure ML for large-scale industrial data analytics.
- Conducted semantic video search via Nvidia VSS and CA-RAG pipelines to review factory floor surveillance footage for safety and operational compliance.
- Built custom embeddings and vector search pipelines using ChromaDB and Elasticsearch to accelerate root cause analysis and fault similarity detection in equipment logs.
- Created ML observability pipelines using WhyLabs, ensuring real-time visibility into model performance, data drift, and latency in production environments.
- Designed interactive executive dashboards in Tableau, Power BI, and Google Data Studio, integrating real-time KPIs for monitoring equipment health and production metrics via REST APIs.
- Enabled low-code/no-code workflows for data scientists and business users while maintaining governance controls.
- Partnered with platform and engineering teams to provision secure, containerized environments for batch and real-time inference.
- Implemented predictive models for equipment failure forecasting, demand surge detection, and capacity planning using XGBoost, LightGBM, and Random Forest algorithms.
- Developed and supported low-latency model scoring APIs for real-time prediction use cases.
- Developed scalable data pipelines using Snowflake to support manufacturing and supply chain analytics workloads.
- Performed data integrity audits and anomaly detection using rule-based and ML techniques to ensure high-quality data for critical analytics workloads.
- Developed RESTful APIs with Flask and FastAPI to expose ML model endpoints for integration with Caterpillar's industrial IoT and network management systems.
- Optimized star and snowflake schemas in Snowflake to enhance BI and predictive analytics capabilities across manufacturing operations.
- Collaborated cross-functionally with data scientists, engineers, and operations teams to align AI models with Caterpillar's productivity and safety KPIs.

Environment: TensorFlow, PyTorch, Keras, Apache Spark, Scala, SQL, Apache Airflow, Prefect, OpenAI GPT-4, LoRA, PEFT, CLIP, BLIP, Nvidia VSS, CA-RAG, ChromaDB, Elasticsearch, WhyLabs, Docker, Kubernetes, SageMaker

Client: Visa, Philadelphia, PA

June 2021 – Sep 2022

Role: Sr Data Science

Responsibilities:

- Extracted and processed patient feedback data using Python, NLTK, and spaCy to perform sentiment analysis and intent classification, enhancing patient experience and care quality analytics for McKesson.
- Developed unsupervised clustering models (K-Means, Hierarchical Clustering, DBSCAN) for patient segmentation to support targeted healthcare programs and personalized treatment outreach.
- Utilized BERT and Word2Vec for embedding generation from unstructured clinical notes and patient reviews, improving text classification and extraction of key health trend insights.
- Processed medical image metadata using OpenCV, integrating quality assurance checks for diagnostic imaging and enabling automated detection of anomalies or defects.
- Built supervised models using SVM, Decision Trees, and Logistic Regression to drive personalized patient content strategies and behavioral health profiling.
- Developed and optimized SQL queries within Snowflake to analyze large-scale health datasets, improving data processing speed by 30% for healthcare decision support.
- Conducted exploratory data analysis (EDA) and implemented dimensionality reduction to address multicollinearity, enhancing interpretability of clinical prediction models.
- Created custom dashboards with Plotly and Dash to visualize NLP-driven health outcome trends for clinical performance review and healthcare service analytics.
- Implemented secure role-based access control (RBAC) and data masking policies in Snowflake to ensure patient data compliance with HIPAA and healthcare governance standards.
- Automated NLP feature engineering using object-oriented Python and optimized clinical text preprocessing with PyTorch for efficient embedding pipeline development.
- Developed RESTful APIs using Flask to support NLP model deployment, validation, and A/B testing for continuous improvement of health informatics applications.
- Implemented CI/CD pipelines using Docker, Jenkins, GitHub Actions, and AWS CodePipeline to ensure seamless deployment of retrained clinical models.
- Monitored deployed health models using AWS CloudWatch and SageMaker Model Monitor, ensuring detection of data drift and anomalies in real-time clinical analytics.
- Queried large-scale healthcare datasets from Google BigQuery and PostgreSQL to support live clinical dashboards and weekly healthcare performance metrics.
- Built visualizations and executive reports using Matplotlib, Seaborn, Power BI, and Tableau to communicate clinical model performance and patient experience insights.
- Applied forecasting techniques using Prophet and time-series models to support medical inventory planning and seasonal demand analysis for pharmaceutical products.
- Engineered pricing optimization models and performed A/B testing to evaluate the impact of healthcare promotional strategies on key clinical and retail KPIs for McKesson.

Environment: Python, NLTK, spaCy, BERT, Word2Vec, OpenCV, K-Means, Hierarchical Clustering, DBSCAN, SVM, Decision Trees, Logistic Regression, EDA, Plotly, Dash, PyTorch, Flask, Docker, Jenkins, GitHub Actions, AWS CodePipeline, AWS CloudWatch

Client: Quotient, Mountain View, CA

Oct 2019 – May 2021

Role: Sr Machine Learning Engineer

Responsibilities:

- Designed and deployed AI-driven systems using Azure Machine Learning, enabling scalable model training, monitoring, and auto-retraining for production use cases in vehicle feedback and service optimization.
- Utilized Azure Cognitive Services for text analytics, speech-to-text, and computer vision to support sentiment mining, call center analytics, and defect detection from inspection images.
- Processed and modeled large-scale vehicle and service datasets using Azure Databricks, PySpark, MLflow, and AutoML for hyperparameter tuning and pipeline orchestration.
- Built predictive models using Python libraries such as scikit-learn, NumPy, pandas, SciPy, Keras, TensorFlow, and PyTorch, applying algorithms like Linear Regression, Logistic Regression, Decision Trees, Random Forest, KNN, SVM, XGBoost, Bagging, Boosting, and Clustering techniques (e.g., K-means, DBSCAN, Gaussian Mixture).
- Developed CNNs using TensorFlow for vehicle image recognition, defect localization, and classification tasks in the quality inspection pipeline.
- Implemented LSTM-based RNNs to build NLP solutions such as intent classification and vehicle feedback categorization from text.
- Automated data ingestion and feature engineering workflows using Python scripts, Airflow, and API integration.
- Leveraged H2O Flow notebooks for model building and performance evaluation in customer satisfaction prediction and failure rate estimation.
- Deployed real-time inference pipelines using Azure Stream Analytics, Azure Event Hubs, and Azure Functions to handle streaming vehicle telemetry and provide insights into driver behavior and anomaly alerts.
- Built microservices using Azure Functions to trigger AI workflows, handle ETL automation, and integrate real-time alerts with operational dashboards.
- Queried operational and telematics data using SQL across Oracle, PostgreSQL, and MySQL, supporting predictive maintenance and warranty analytics.
- Used PySpark for distributed data processing on large datasets (>10M rows) in big data environments.
- Wrote modular, testable, and reusable code using object-oriented programming (OOP) and Pythonic practices.
- Used Apache Zeppelin notebooks with Apache Spark for unified big data exploration, model prototyping, and real-time insights on service activity.
- Applied Spark Streaming for real-time sentiment classification, alert generation, and service-level adjustments using live feedback data.
- Performed A/B testing and statistical analysis using SciPy, statsmodels, and bootstrapping methods.
- Designed data visualizations and dashboards using Tableau, Power BI, and R Studio to communicate model performance and incident trends to business stakeholders.

Environment: Azure Machine Learning, Azure Cognitive Services, Azure Databricks, AutoML, PySpark, MLflow, Keras, TensorFlow, PyTorch, scikit-learn, NumPy, pandas, SciPy, CNN, LSTM, RNN, H2O Flow, XGBoost.

Client: TD Bank, Chicago, IL

Sep 2017-Sep 2019

Role: Data Analyst

Responsibilities

- Collected, cleaned, and integrated structured and semi-structured data from Oracle databases, internal systems, and customer support platforms to create reliable analytics datasets.
- Performed data preprocessing, validation, and exploratory data analysis (EDA) using Python to identify trends, patterns, and anomalies.
- Analyzed large datasets using Pandas, NumPy, and SciPy to support data-driven decision-making across business functions.

- Built descriptive and diagnostic analyses to evaluate customer behavior, product performance, and operational efficiency.
- Applied statistical methods and basic predictive techniques (e.g., logistic regression, clustering) to support forecasting and segmentation initiatives.
- Conducted text and sentiment analysis on customer feedback and support data to uncover insights into customer satisfaction and service quality.
- Designed and maintained automated ETL workflows using Python and Apache Airflow to ensure data accuracy, consistency, and timely availability.
- Queried and optimized complex datasets using SQL across Oracle, PostgreSQL, and Cassandra to support reporting and ad hoc analysis requests.
- Developed interactive dashboards and visual reports using Tableau to communicate KPIs, trends, and insights to stakeholders.
- Integrated external data sources through web scraping and APIs to enrich analytical models and reporting datasets.
- Collaborated with cross-functional teams to translate business requirements into analytical solutions and actionable insights.

Environment: Python, Pandas, NumPy, SciPy, Seaborn, Matplotlib, SQL, Oracle, PostgreSQL, Cassandra, Apache Airflow, Tableau, AWS, Azure Blob Storage, Azure SQL Database, Cosmos DB

Client: WNS Global Services, Bengaluru, India

Aug 2013 - June 2016

Role: Data Analyst

Responsibilities

- Collected, validated, and analyzed large operational and customer datasets to support business reporting and performance analysis for global clients.
- Performed data cleaning, transformation, and aggregation using SQL and Excel to ensure accuracy and consistency of reporting datasets.
- Developed recurring and ad hoc reports to track key operational KPIs, service-level metrics, and process efficiency indicators.
- Conducted exploratory data analysis to identify trends, root causes, and improvement opportunities in business operations.
- Created dashboards and visual reports using Tableau and Excel to communicate insights to stakeholders and senior management.
- Worked closely with operations, quality, and client-facing teams to translate business requirements into analytical deliverables.
- Automated manual reporting processes using Python and SQL, improving report turnaround time and reducing errors.
- Supported forecasting and capacity planning activities through historical data analysis and trend modeling.
- Ensured data governance and compliance by following data quality standards and documentation practices.
- Assisted in client presentations by summarizing analytical findings and providing actionable recommendations.